

solution system of linear equations

حل نظام المعادلات الخطية

one solution
حل وحيد

بعد التبسيط

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = b_1$$

$$c_1 x_2 + c_2 x_3 = b_2$$

$$m_1 x_3 = b_3$$

unique solution
أكثر من حل

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = b_1$$

$$c_1 x_2 + c_2 x_3 = b_2$$

$$0 = 0$$

عدد المعادلات بعد التبسيط

أقل من عدد الجاهيل

No solution
لا يوجد حل

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = b_1$$

$$a_2 x_2 + a_3 x_3 = b_2$$

$$0 = b_3$$

solve the system (by using Gauss)

$$x + 2y - 3z = 6 \quad L_1$$

$$2x - y + 4z = 2 \quad L_2$$

$$4x + 2y - 2z = 14 \quad L_3$$

sol

$$x + 2y - 3z = 6$$

$$-5y + 10z = -10 \quad L_2 \rightarrow -2L_1 + L_2$$

$$-5y + 10z = -10 \quad L_3 \rightarrow -4L_1 + L_3$$

$$x + 2y - 3z = 6$$

$$-5y + 10z = -10$$

$$0 = 0$$

$$-5y = -10 - 10z$$

$$5y = 10 + 10z$$

$$y = 2 + 2z$$

let $z = 1$

$$y = 2 + 2(1)$$

$$y = 2 + 2 = 4$$

$$L_3 \rightarrow -L_2 + L_3$$

$$y = 4$$

$$x + 2(4) - 3(1) = 6$$

$$x + 8 - 3 = 6$$

$$x = 6 - 5 = 1$$

$$x = 1$$

$$u = (1, 4, 1) \text{ solution}$$

$$z = a \quad y = 2 + 2a$$

$$x + 2(2 + 2a) - 3a = 6$$

$$x + 4 + 4a - 3a = 6$$

$$x = 2 - a$$

$$u(2 - a, 2 + 2a, a)$$

solve the system

$$\begin{aligned} x + 2y - 3z &= -1 \\ 3x - y - 2z &= 7 \\ 5x + 3y - 4z &= 2 \end{aligned} \quad \begin{matrix} L_1 \\ L_2 \\ L_3 \end{matrix}$$

sd

$$\begin{aligned} x + 2y - 3z &= -1 \\ -7y + 7z &= 10 & L_2 \rightarrow -3L_1 + L_2 \\ -7y + 11z &= 7 & L_3 \rightarrow -5L_1 + L_3 \end{aligned}$$

$$\frac{31}{28} - \frac{122}{28} + \frac{9}{4} = \frac{-91}{28} + \frac{9(7)}{4(7)} = \frac{-28}{28} = -1$$

$$x + 2y - 3z = -1$$

$$-7y + 7z = 10$$

$$4z = -3$$

$$z = \frac{-3}{4}$$

$$-7y + 7\left(\frac{-3}{4}\right) = 10$$

$$-7y - \frac{21}{4} = 10$$

$$-28y - 21 = 40$$

$$L_3 \rightarrow -L_2 + L_3$$

$$-2y = 40 + 21$$

$$-28y = 61$$

$$y = \frac{61}{-28} = -\frac{61}{28}$$

$$y = -\frac{61}{28}$$

$$x + 2\left(-\frac{61}{28}\right) - 3\left(\frac{-3}{4}\right) = -1$$

$$x - \frac{61}{14} + \frac{9}{4} = -1$$

(28)

$$x = \frac{31}{28}$$

solve the system

$$x + 2y - 3z = -1$$

$$3x - y + 2z = 7$$

$$5x + 3y - 4z = 2$$

L_1
 L_2
 L_3

sol

$$x + 2y - 3z = -1$$

$$-7y + 11z = 10$$

$$-7y + 11z = 7$$

$$L_2 \rightarrow -3L_1 + L_2$$

$$L_3 \rightarrow -5L_1 + L_3$$

$$x + 2y - 3z = -1$$

$$-7y + 11z = 10 \quad L_3 \rightarrow -L_2 + L_3$$

$$0 + 0 = -3$$

$$0 = -3$$

no solution

ليس حل

solve the system

$$\begin{array}{lcl} x + 2y - 3z = 4 & L_1 & \\ x + 3y + z = 11 & L_2 & \\ 2x + 5y - 4z = 13 & L_3 & \end{array}$$

sol

$$x + 2y - 3z = 4$$

$$y + 4z = 7 \quad L_2 \rightarrow -L_1 + L_2$$

$$y + 2z = 5 \quad L_3 \rightarrow -2L_1 + L_3$$

$$x + 2y - 3z = 4$$

$$y + 4z = 7$$

$$-2z = -2 \quad \times -2$$

$$z = 1$$

$$y + 4(1) = 7$$

$$y = 7 - 4$$

$$y = 3$$

$$x + 2(3) - 3(1) = 4$$

$$x + 6 - 3 = 4$$

$$x = 4 - 3$$

$$x = 1$$

$$L_3 \rightarrow -L_2 + L_3$$

$$u = (x, y, z)$$

$$u = (1, 3, 1)$$

solution of Homogeneous of Linear Equation

solve the homogeneous system

$$x + 2y - 3z + w = 0 \quad L_1$$

$$x - 3y + z - 2w = 0 \quad L_2$$

$$2x + y - 3z + 5w = 0 \quad L_3$$

sol

$$x + 2y - 3z + w = 0$$

$$-5y + 4z - 3w = 0 \quad L_2 \rightarrow -L_1 + L_2$$

$$-3y + 3z + 3w = 0 \quad L_3 \rightarrow -2L_1 + L_3$$

$$\begin{aligned} 15y - 12z + 9w &= 0 \\ -15y + 15z + 15w &= 0 \end{aligned}$$

$$x + 2y - 3z = 0 \quad w=0$$

$$-5y + 4z - 3w = 0$$

$$3z + 24w = 0$$

$$3z = -24w$$

$$z = -8w$$

$$\text{let } w = 1$$

$$z = -8$$

$$L_3 \rightarrow -3L_2 + 5L_3$$

$$u = (x, y, z, w)$$

$$u = (-11, -7, -8, 1)$$

$$-5y + 4(-8) - 3(1) = 0$$

$$-5y - 32 - 3 = 0$$

$$-5y = 35 \quad \therefore -5$$

$$y = -7$$

$$x + 2(-7) - 3(-8) + 1 = 0$$

$$x - 14 + 24 + 1 = 0$$

$$x = -11$$

solution of Homogeneous of Linear Equation

solve the homogeneous system

$$x + y - z = 0$$

$$2x + 4y - z = 0$$

$$3x + 2y + 2z = 0$$

$$\begin{array}{r} -3y \\ 4y \end{array}$$

sol

$$x + y - z = 0$$

$$2y + z = 0$$

$$-y + 5z = 0$$

$$L_2 \rightarrow -2L_1 + L_2$$

$$L_3 \rightarrow -3L_1 + L_3$$

$$x + y - z = 0$$

$$-y + 5z = 0$$

$$2y + z = 0$$

$$x + y - z = 0$$

$$-y + 5z = 0$$

$$11z = 0$$

$$L_3 \rightarrow 2L_2$$

$$\boxed{z = 0}$$

$$-y + 0 = 0$$

$$\boxed{y = 0}$$

$$x + 0 + 0 = 0$$

$$\boxed{x = 0}$$

solution of Homogeneous of Linear Equation

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$$

$$c_1 x_2 + c_2 x_3 = 0$$

$$m x_3 = 0$$

$$u (0, 0, 0)$$

عدد المعادلات $r = n$

the the system has only the zero solution
حل صفري

عدد
المعادلات

$$r < n$$

unique solution

solve the system (by using Gauss)

$$x + 2y - 3z = 4 \quad L_1$$

$$x + 3y + z = 11 \quad L_2$$

$$2x + 5y - 4z = 13 \quad L_3$$

$$2x + 6y + 2z = 22 \quad L_4$$

sol

$$x + 2y - 3z = 4$$

$$y + 4z = 7 \quad L_2 \rightarrow -L_1 + L_2$$

$$y + 2z = 5 \quad L_3 \rightarrow -2L_1 + L_3$$

$$2y + 8z = 14 \quad L_4 \rightarrow -2L_1 + L_4$$

$$x + 2y - 3z = 4$$

$$y + 4z = 7$$

$$-2z = -2$$

$$0 = 6$$

متضاد

شكل

نفس المتال لاول

$$L_3 \rightarrow -L_2 + L_3$$

$$L_4 \rightarrow -2L_2 + L_4$$

T or F

If $r=n$ then is there are as

many equations as unknowns

then the system has a unique solution

()

(F)

solve

$$\begin{cases} 2x + y - 2z + 3w = 1 \\ 3x + 2y - z + 2w = 4 \\ 3x + 3y + 3z - 3w = 5 \end{cases}$$

sol

$$\begin{cases} 2x + y - 2z + 3w = 1 \\ y + 4z - 5w = 5 \\ 3y + 12z - 15w = 7 \end{cases}$$

$$\begin{cases} -6x - 3y + 6z - 9w = -3 \\ 6x + 4y - 2z + 4w = 8 \\ -6x - 3y + 6z - 9w = -3 \\ 6x + 6y + 6z - 6w = 10 \end{cases}$$

$$L_2 \rightarrow -3L_1 + 2L_2$$

$$L_3 \rightarrow -3L_1 + 2L_3$$

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$0 + 0 + 0 = 8$$

$$0 = 8$$

no solution

$$L_3 \rightarrow -3L_2 + L_3$$

solve

$$\begin{aligned} 2x + y - 2z + 3w &= 1 \\ 3x + 2y - z + 2w &= 4 \\ 3x + 3y + 3z - 3w &= 5 \end{aligned}$$

sol

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$3y + 12z - 15w = 7$$

$$\begin{aligned} -6x - 3y + 6z - 9w &= -3 \\ 6x + 4y - 2z + 4w &= 8 \\ -6x - 3y + 6z - 9w &= -3 \\ -6x + 6y + 6z - 6w &= 10 \end{aligned}$$

$$L_2 \rightarrow -3L_1 + 2L_2$$

$$L_3 \rightarrow -3L_1 + 2L_3$$

$$2x + y - 2z + 3w = 1$$

$$y + 4z - 5w = 5$$

$$0 + 0 + 0 = 8$$

$$L_3 \rightarrow -3L_2 + L_3$$

$$0 = 8$$

no solution

solution of Homogeneous of Linear Equation

solve the homogeneous system

$$x + y - z = 0$$

$$2x - 3y + z = 0$$

$$x - 4y + 2z = 0$$

sol

$$x + y - z = 0$$

$$L_2 \rightarrow -2L_1 + L_2$$

$$-5y + 3z = 0$$

$$L_3 \rightarrow -L_1 + L_3$$

$$-5y + 3z = 0$$

$$x + y - z = 0$$

$$-5y + 3z = 0$$

$$0 = 0$$

$$3z = 5y$$

$$z = \frac{5}{3}y$$

$$\text{Let } y = 3$$

$$z = \frac{5}{3}(3)$$

$$z = 5$$

$$L_3 \rightarrow -L_2 + L_3$$

$$x - 3 - 5 = 0$$

$$x = 2$$

$$u = (2, 3, 5)$$